

FMS QA Verification Test Matrix

Scope: Date-Based Scheduling, Double-Assignment Prevention, Driver Scores, and Vehicle Health Caching.

ID	Feature / Area	Test Case Goal	Steps	Expected Result
TC-1.1	Scheduling & Overlaps	Driver Double-Assignment Block (Manual)	1. Create a trip (Trip A) from 10:00 AM to 12:00 PM assigned to Driver X. 2. Try to create/edit Trip B from 11:00 AM to 1:00 PM. 3. Select Driver X in the picker.	UI blocks the assignment, displays a friendly warning, and prevents creation. The database trigger aborts the save if the UI is bypassed.
TC-1.2	Scheduling & Overlaps	Vehicle Double-Assignment Block (Manual)	1. Assign Vehicle Y to Trip A (10:00 AM - 12:00 PM). 2. Try to assign Vehicle Y to Trip B (11:00 AM - 1:00 PM).	Vehicle selection filters out Vehicle Y or shows a validation warning that the vehicle is already booked.
TC-1.3	Scheduling & Overlaps	Driver Unavailability Blockout	1. Add schedule blockout (is_available = false) for Driver X from 2:00 PM to 4:00 PM. 2. Try to assign Driver X to a trip from 2:30 PM to 3:30 PM.	Driver is filtered out from the eligible list; UI prevents creation with a schedule conflict error.
TC-1.4	Scheduling & Overlaps	Non-Overlapping Scheduling (Success)	1. Assign Driver X & Vehicle Y to a trip from 8:00 AM to 9:30 AM. 2. Assign them to a trip from 10:00 AM to 11:30 AM (no overlap).	Both trips are created successfully without warnings.
TC-2.1	Driver Scores	Optimized Score Query	1. Complete a trip with multiple deviation alerts. 2. Recalculate driver score.	Deviation alert query executes as a single batched .in(...) statement instead of a loop query.
TC-2.2	Driver Scores	Missing Scores Warning Banner	1. Ensure active completed trips exist but no scores exist in the DB. 2. Navigate to Driver Performance reports.	Orange warning banner displays at the top, advising the manager to calculate scores.
TC-2.3	Driver Scores	Exporter N/A Fallback	1. Export PDF/CSV Driver Performance report. 2. Locate a driver who does not have a calculated score yet.	Overall Score column displays N/A instead of a static fallback score of 75.
TC-3.1	Vehicle Health	Real Health Score Display	1. Navigate to Vehicles list view. 2. Check Health capsules.	Capsules display calculated values (0-100%) rather than static status-based mocks (e.g. always 92 or 42).
TC-3.2	Vehicle Health	24-Hour Cache Expiration	1. Load scores (instant cache load). 2. Modify calculated_at of cache in DB to be > 24 hours old. 3. Reload screen.	App detects expired cache, runs background calculation, updates DB cache, and displays the new score.
TC-3.3	Vehicle Health	Duplicate Records Crash Mitigation	1. Insert duplicate vehicle record (same VIN/ID) in DB list. 2. Open Manager vehicles view.	Screen loads successfully and merges duplicates without triggering a runtime uniqueKeys mapping crash.